1.3 Using Midpoint and Distance Formulas

Name _____ Period ____ Date _____

Geometry 1.3 (30-minute plan) · Student sheet

I can…

- use a midpoint or segment bisector to find lengths.
- use the Midpoint Formula to find a midpoint or an endpoint.
- use the Distance Formula to find the length of a segment.

Vocabulary · fill in the missing word

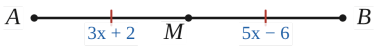
Midpoint — the point that divides a segment into two _____ segments

Segment bisector — a point, ray, line, segment, or plane that _____ a segment at its midpoint

Part 1 · Notes and examples (fill in each note, then work the example)

Example 1

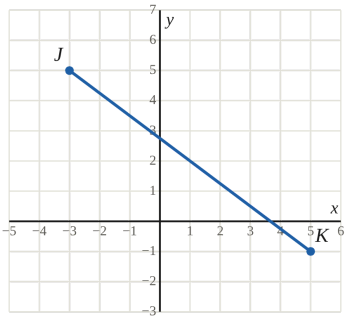
M is the midpoint of $\overline{AB}$. AM = 3x + 2 and MB = 5x − 6. Find x and AB.



Midpoint Formula: M = (_____, _____)

Example 2

Find the midpoint of $\overline{JK}$ with J(−3, 5) and K(5, −1).

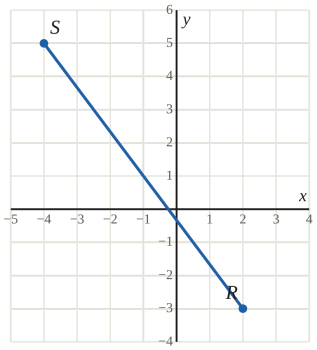


Finding an endpoint: the midpoint is _____ between the endpoints, so add the same change again.

Distance Formula: AB = _____

Example 3

Find the distance between R(2, −3) and S(−4, 5).



Part 2 · Partner practice

Partner practice

Find the distance between (1, −2) and (6, 10).